

The question of declining sperm density revisited: an analysis of 101 studies published 1934-1996.

In 1992 Carlsen et al. reported a significant global decline in sperm density between 1938 and 1990 [Evidence for Decreasing Quality of Semen during Last 50 Years. *Br Med J* 305:609-613 (1992)]. We subsequently published a reanalysis of the studies included by Carlsen et al. [Swan et al. Have Sperm Densities Declined? A Reanalysis of Global Trend Data. *Environ Health Perspect* 105:1228-1232 (1997)]. In that analysis we found significant declines in sperm density in the United States and Europe/Australia after controlling for abstinence time, age, percent of men with proven fertility, and specimen collection method. The declines in sperm density in the United States (approximately 1.5%/year) and Europe/Australia (approximately 3%/year) were somewhat greater than the average decline reported by Carlsen et al. (approximately 1%/year). However, we found no decline in sperm density in non-Western countries, for which data were very limited. In the current study, we used similar methods to analyze an expanded set of studies. We added 47 English language studies published in 1934-1996 to those we had analyzed previously. The average decline in sperm count was virtually unchanged from that reported previously by Carlsen et al. (slope = -0.94 vs. -0.93). The slopes in the three geographic groupings were also similar to those we reported earlier. In North America, the slope was somewhat less than the slope we had found for the United States (slope = -0.80; 95% confidence interval (CI), -1.37--0.24). Similarly, the decline in Europe (slope = -2.35; CI, -3.66--1.05) was somewhat less than reported previously. As before, studies from other countries showed no trend (slope = -0.21; CI, -2.30-1.88). These results are consistent with those of Carlsen et al. and our previous results, suggesting that the reported trends are not dependent on the particular studies included by Carlsen et al. and that the observed trends previously reported for 1938-1990 are also seen in data from 1934-1996.